# Ryan P. Seaman

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## SUMMARY

Biomedical informatics researcher and MMSc candidate at Harvard Medical School with experience in collaborative data analysis, research visualization, and scientific training. Interested in research and project roles connecting biomedical science with useful health technology.

## EXPERIENCE

**Graduate Research Assistant, HIDIVE Lab** Dec 2025 - Present
Harvard Medical School | Boston, MA

- Developing VitessceGen, a multi-agent AI tool that generates interactive visualization configurations from spatial and single-cell biomedical data.
- Evaluating generated configurations against expert-authored examples using HuBMAP datasets, under Nils Gehlenborg's supervision.

**Summer Research Intern, HIDIVE Lab** May 2026 - Aug 2026
Harvard Medical School | Boston, MA

- Created the full annotation and narrative system within the existing Vitessce framework, with supervision and guidance from Mark S. Keller and Nils Gehlenborg.
- Connected narrative text, saved views, and visual annotations into guided stories for spatial and single-cell data; manuscript in preparation.

**Bioinformatician** Jan 2023 - Aug 2025
MDI Biological Laboratory, Comparative Genomic and Data Science Core | Bar Harbor, ME

- Collaborated with research labs on biomedical data analysis and developed tools that help researchers explore their own datasets.
- Built scscape, an open-source Nextflow pipeline for reproducible, multi-sample single-cell RNA sequencing analysis.
- Maintained GCP/AWS infrastructure for research analysis and data storage.

## EDUCATION

**Harvard Medical School** Expected May 2027
MMSc, Biomedical Informatics | Boston, MA

**Colby College** Jan 2023
BA, Computational Biology | Waterville, ME | Varsity soccer, four years

## TEACHING & MENTORING

**MDI Biological Laboratory** 2023 - 2025

- Co-taught a three-day single-cell course in 2024; created teaching materials and the course website.
- Taught Python, held office hours, and mentored summer interns on computational research projects.

## TECHNICAL SKILLS

Python, R; biomedical data analysis and visualization; single-cell and spatial genomics.

## SELECTED PUBLICATION

**Seaman RP**, et al. (2024). A cloud-based training module for efficient de novo transcriptome assembly using Nextflow and Google Cloud. Briefings in Bioinformatics.